

Project Title : Exploratory Data Analysis

Project Name : Sample Superstore Sales Analysis

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Submission Date : 8/7/2026

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Business Problem :

The company aims to understand the factors affecting profitability across different product categories, regions, customer segments, and discount levels. Although the business generates sales across these dimensions, profitability is not consistent, making it important to identify the areas contributing to lower profit.

Profitability is a key business metric that supports effective decision-making, pricing strategies, and long-term financial performance. Identifying the factors associated with low profitability enables the company to make informed decisions and improve its overall financial outcomes.

This report investigates the key factors influencing profitability by analyzing sales, profit, product categories, regions, customer segments, and discount levels. The objective of this analysis is to identify the major areas associated with lower profitability and provide data-driven insights that can support future business decisions.

Business Objective :

- To identify factors affecting profitability
- To analyze profitability of categories and sub-categories
- To investigate impact of discounts on profitability
- To analyze profitability across different regions
- To evaluate segment contribution to profitability
- To examine which ship mode is profitable and which not

Dataset Overview :

The dataset which is used for investigating the business problem is SampleSuperstore which is downloaded from Kaggle platform. The dataset belongs to the Retail Sales business domain. This dataset has 9994 records and 13 features in it. The dataset consists of features which are profit, sales, discount values, segments, product categories, regions, shipping modes and also the customer and geographic information. The reason behind choosing this dataset is that it contains the variable required to analyze profitability across multiple business dimensions

Data Cleaning :

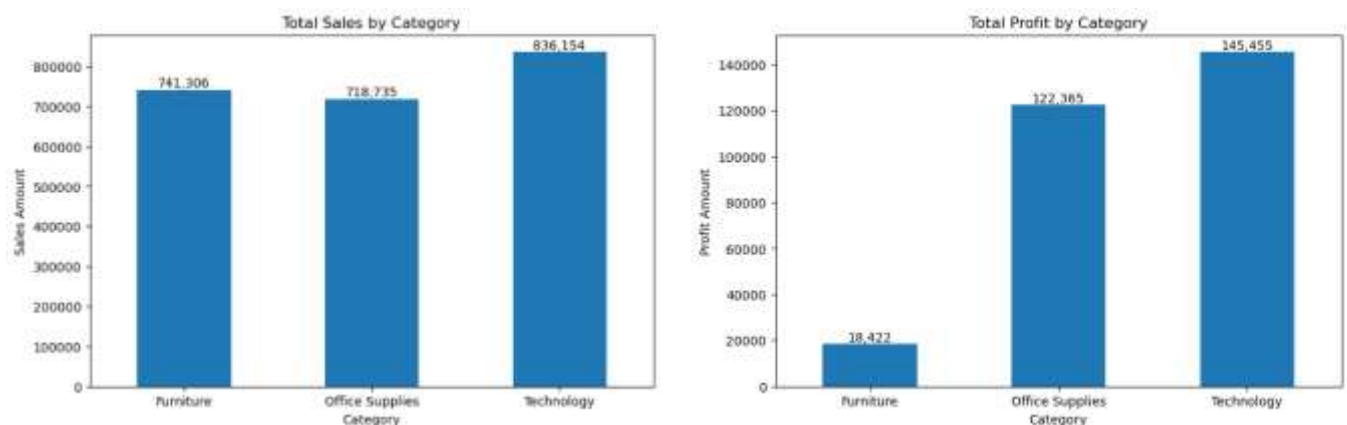
Data Quality Check	Observation	Action Taken
Missing Values	No missing values	No action required
Duplicate Records	17 Duplicate records	Removed duplicates
Data Types	Appropriate	No action required
Outliers	Genuine business transaction	Retained for analysis

5. Exploratory Data Analysis

5.1 Category Performance Analysis :

i) Business Question : Which is the highest contributing category in terms of Sales and Profit and which require business attention

ii) Visualization:



iii) Observation : Investigation states that the Category Technology generated the highest sales and profit among all product categories. While the Furniture is a category which is second-highest sales contributing category but at the same time it significantly less profit generating category. Although the least sales generating category is Office Supplies still its profit generation capability is better than Furniture

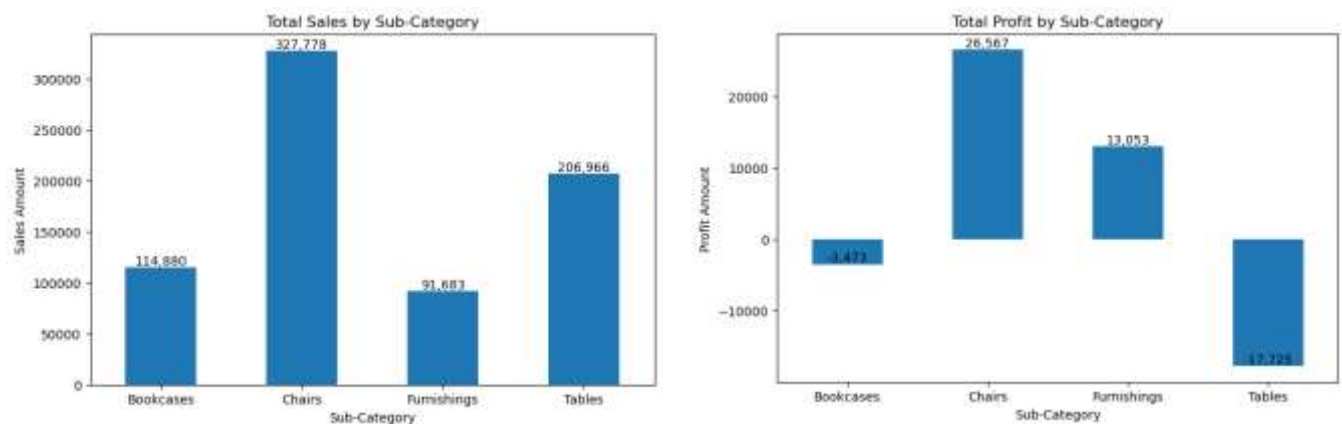
iv) Key findings: Technology generated the highest sales and profit among all product categories. Furniture generated high sales but comparatively low profitability, indicating margin-related issues.

v) Business Insight: This suggests that generating highest sales is insufficient but improving profit margin is also equally important within furniture category should be the priority

5.2 Sub-category Performance Analysis :

i) Business Question : Which Sub-Category of Furniture is responsible for lower profit ?

ii) Visualization :



iii) Observation : This visualization answers that Chairs generated the large amount of sales and also the profit comparatively. On the other hand the Sub-Category Tables generated second-highest sales and has the negative profit numbers . Bookcases category's profit number is significantly less negative than Tables

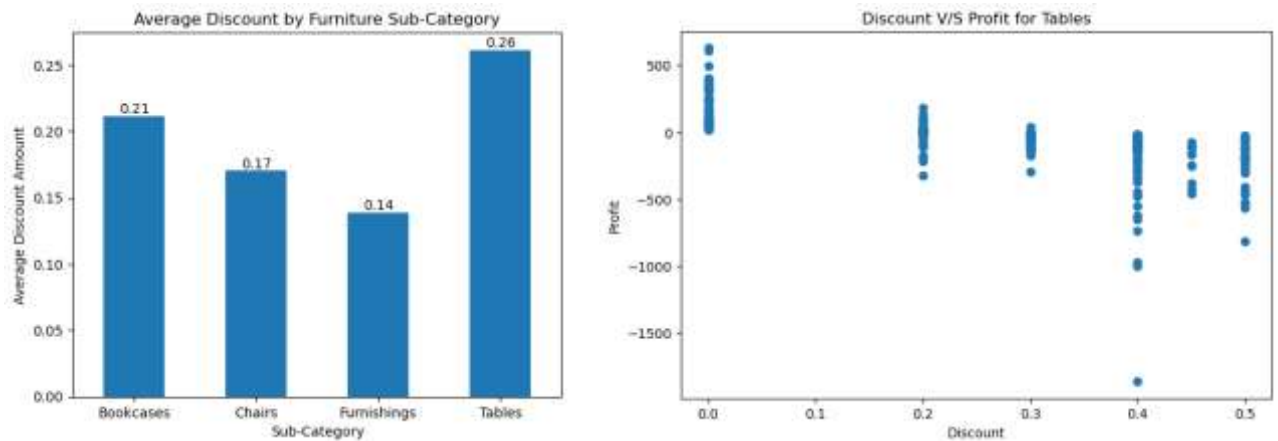
iv) Key findings: Within the Furniture category, Tables recorded the highest losses despite generating substantial sales.

v) Business Insight : The negative profitability in Tables categories should be analyzed further to address the main cause of profit droppage

5.3 Discount Impact Analysis :

i) Business Question : Do Higher Discounts reduce profitability?

ii) Visualization:



iii) Observation : On the basis of visualization and statistical analysis we can observe that the range of 40-50% of discount has significantly more negative profit numbers while the range of 20-30% discount has balanced negative and positive values. Where the discount percentage is 0-10% the profits are rising . this analysis supports the hypothesis that, discounts and profits are correlated.

iv) Key findings: Tables had the highest average discount. Higher discounts were associated with lower profit.

iv) Business Insight: Discounts are affecting the profitability and are one of those reasons who are affecting the profitability of the business. Need to investigate for other features too

5.4 Regional Performance Analysis

- i) **Business Question** : Which region contributes the highest and the lowest profitability?
- ii) **Observation** : The regional profit analysis reveals noticeable differences in profitability across regions. The West region generated the highest overall profit among all regions. In contrast, the East region recorded the lowest profitability, followed by the South region, while the Central region also showed comparatively lower profit than the West.
- iii) **Key Findings** : Profitability varies significantly across different regions, with the East region requiring immediate business attention.
- iv) **Business Insight** : The findings indicate that geographical location has a considerable influence on profitability. The company should further investigate factors such as product demand, pricing strategies, discount policies, and operational costs in the East and South regions to improve their financial performance.

5.5 Customer Segment Performance Analysis

- i) **Business Question** : Which customer segment contributes the highest and the lowest profitability?
- ii) **Observation** : The analysis shows that profitability differs across customer segments. The Consumer segment generated the highest overall profit, followed by the Corporate segment. The Home Office segment contributed the lowest profit among all customer segments.
- iii) **Key Findings** : Customer profitability is not equally distributed, with the Home Office segment generating comparatively lower profit.
- iv) **Business Insight** : This suggests that customer segments respond differently to the company's products and pricing strategies. Understanding the purchasing behavior and profitability of the Home Office segment may help identify opportunities to improve its financial contribution.

5.6 Shipping Mode Performance Analysis

i) Business Question : Which shipping mode contributes the highest and the lowest profitability?

ii) Observation : The shipping mode analysis indicates clear differences in profitability across delivery methods. Standard Class generated the highest overall profit, followed by Second Class and First Class. Same Day shipping recorded the lowest profitability among all shipping modes.

iii) Key Findings : Shipping mode has a measurable impact on profitability, with Same Day shipping contributing the least profit.

iv) Business Insight : The findings suggest that faster shipping options may involve higher operational costs or lower profit margins. The company should evaluate shipping costs, pricing strategies, and customer demand to optimize the profitability of different shipping modes.

6. Overall Key Findings :

- Technology generated the highest sales and profit among all product categories.
- Furniture generated high sales but comparatively low profitability, indicating margin-related issues.
- Within the Furniture category, Tables recorded the highest losses despite generating substantial sales.
- Tables also received the highest average discount, suggesting a strong association between discounting and reduced profitability.
- Scatter plot, correlation analysis, and hypothesis testing supported a negative relationship between discounts and profit.
- The West region achieved the highest profitability, whereas the East region recorded the lowest.
- The Consumer segment generated the highest profit, while the Home Office segment contributed the least.
- Standard Class was the most profitable shipping mode, whereas Same Day shipping generated the lowest profit.

7. Business Recommendations :

- Review the pricing and discount strategy for the Tables sub-category to improve profitability.
- Analyze the cost structure of Furniture products to identify additional factors affecting profit margins.
- Evaluate discount policies to ensure they balance customer acquisition with sustainable profitability.
- Investigate the reasons for low profitability in the East region and implement region-specific improvement strategies.
- Develop targeted business strategies for the Home Office customer segment to enhance its profitability.
- Review the operational efficiency and pricing of Same Day shipping to reduce associated costs.
- Continuously monitor profitability across categories, regions, customer segments, and shipping modes to support data-driven business decisions.

8. Conclusion :

The exploratory data analysis identified several factors influencing profitability across different business dimensions. Although some product categories generated strong sales, profitability was not consistent across all categories, regions, customer segments, and shipping modes. The analysis highlighted the Furniture category, particularly the Tables sub-category, as a major area requiring business attention due to its low profitability and high discount levels. Overall, the findings provide data-driven insights that can help the company optimize pricing strategies, improve operational performance, and make informed decisions to enhance long-term profitability.